

SIMATS ENGINEERING
ASSESSMENT TOOL II — DATA ANALYSIS PROJECT

Course: Data Handling and Visualization	Code: DSA0603	CO: CO3
Duration: 180 min	Total Marks: 100	Reg No : 192324298 Name: Akash S

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

Code of Conduct:

I Akash S 192324298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Understanding of Concepts 25%	Analytical Design 25%	Use of Tools 20%	Analysis of Results 20%	Documentation 10%	Total
Q1						
Q2						
Q3						
Q4						
Q5						

DSA0603 — DATA HANDLING AND VISUALIZATION

CO3_AT2_Assessment Tool: DATA ANALYSIS PROJECT

Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by-Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

General Instruction: For each project task, use an appropriate visualization in R, justify the design choice, interpret the result in context, and identify potential visualization bias. R/ggplot2 (or an equivalent tool such as treemapify, ggalluvial, or networkD3) may be used. Submit R code, output plots, and a short written interpretation for each question.

Question 1 — Comparing Distributions with ECDF and Q-Q Plots

Scenario: A logistics company records delivery time (hours) for two courier partners over 20 shipments each. Operations wants to know whether one partner is consistently faster and whether either partner's delivery times follow a normal pattern that would justify simple mean-based SLAs.

Dataset:

Partner	Delivery Times (hours), n = 20
Partner A	18, 19, 20, 20, 21, 21, 22, 22, 23, 23, 24, 24, 25, 26, 27, 28, 30, 33, 38, 46
Partner B	20, 21, 22, 22, 23, 23, 23, 24, 24, 24, 24, 25, 25, 25, 26, 26, 27, 27, 28, 29

Tasks:

1. Construct overlaid ECDF plots for both partners on the same axes.
2. Construct a normal Q-Q plot for each partner separately.
3. Using the ECDF, determine at what delivery-time threshold Partner A and Partner B diverge most, and interpret this in operational terms.
4. Using the Q-Q plots, assess whether either partner's data supports a normal-distribution assumption; identify evidence of skew or outliers.
5. Recommend whether SLA limits should be set using mean $\pm$ standard deviation or empirical percentiles, with justification.

Key competency: ECDF, Q-Q plot, distribution comparison, skewness, outliers

Task 1: Overlaid ECDF Plots (R Code & Description)

R Code for Overlaid ECDF Plots

```
library(ggplot2)
```

```
df_q1 <- data.frame(
  DeliveryTime = c(18,19,20,20,21,21,22,22,23,23,24,24,25,26,27,28,30,33,38,46,
    20,21,22,22,23,23,23,24,24,24,24,25,25,25,26,26,27,27,28,29),
  Partner = rep(c("Partner A", "Partner B"), each = 20)
)

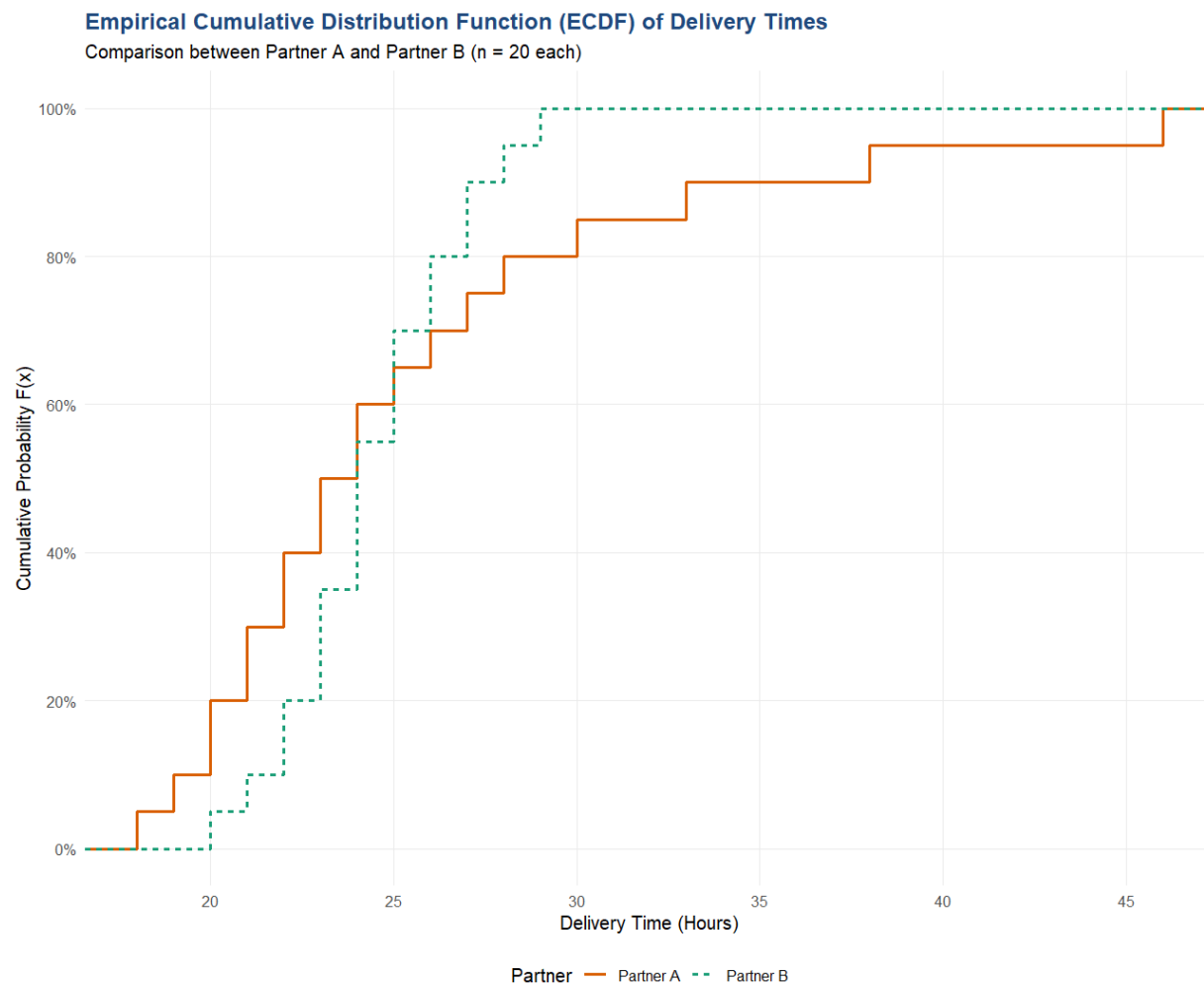
ggplot(df_q1, aes(x = DeliveryTime, color = Partner, linetype = Partner)) +
  stat_ecdf(geom = "step", size = 1.1) +
  scale_color_manual(values = c("Partner A" = "#D95F02", "Partner B" = "#1B9E77")) +
  scale_x_continuous(breaks = seq(15, 50, by = 5)) +
  scale_y_continuous(breaks = seq(0, 1, by = 0.2), labels = scales::percent) +
  labs(title = "Empirical Cumulative Distribution Function (ECDF) of Delivery Times",
```

```

    subtitle = "Comparison between Partner A and Partner B (n = 20 each)",
    x = "Delivery Time (Hours)", y = "Cumulative Probability F(x)" +
    theme_minimal(base_size = 12) +
    theme(legend.position = "bottom",
          panel.grid.minor = element_blank(),
          plot.title = element_text(face = "bold", color = "#1F497D"))

```

Interpretation of Visualization: The ECDF curves track the exact step-wise cumulative probability for each partner without introducing any binning bias or bandwidth smoothing assumptions.



Task 2: Normal Q-Q Plots for Both Partners (R Code & Output Assessment)

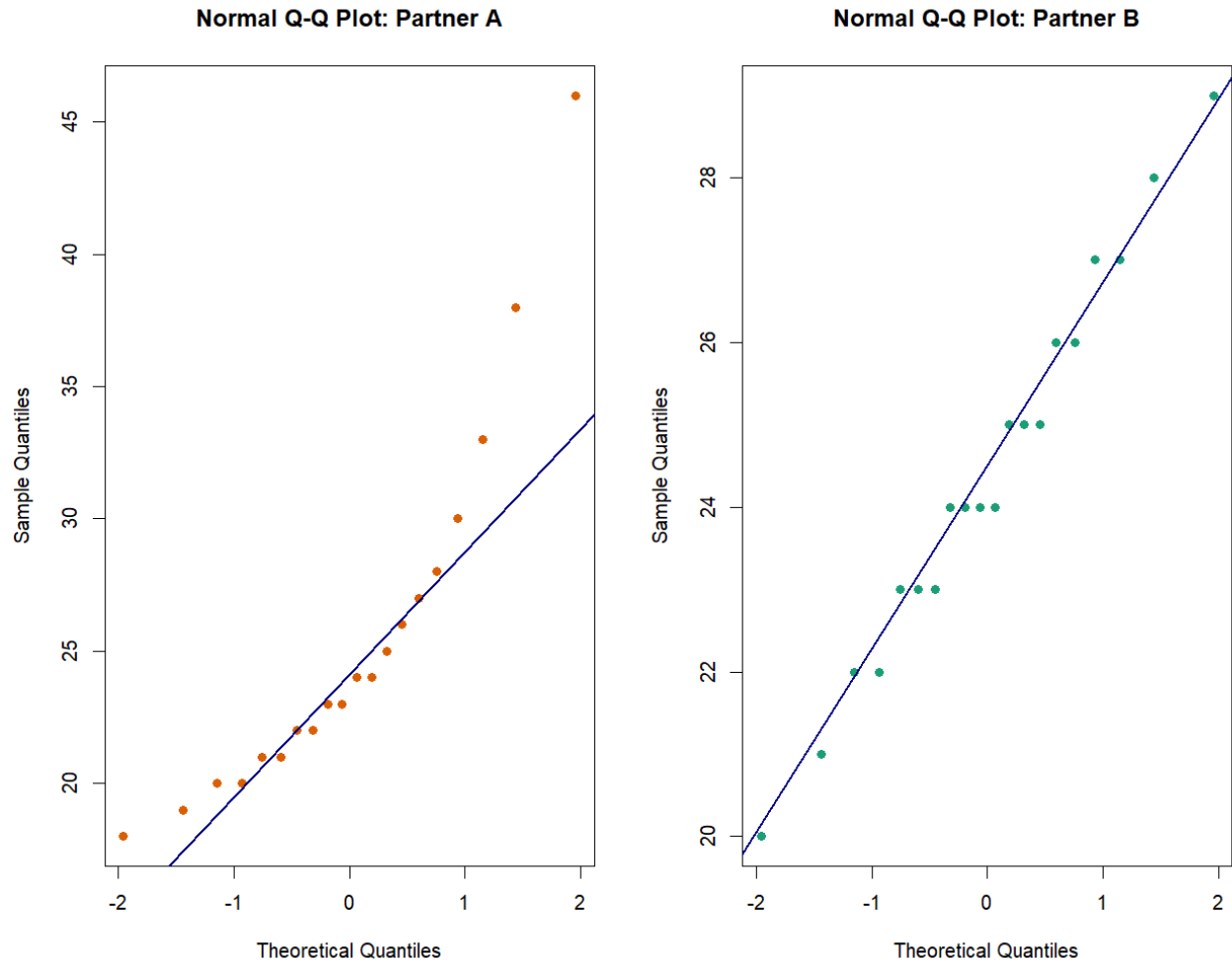
```

# R Code for Q-Q Plots
par(mfrow = c(1, 2))
# Partner A Q-Q Plot
qqnorm(df_q1$DeliveryTime[df_q1$Partner == "Partner A"],
       main = "Normal Q-Q Plot: Partner A", col = "#D95F02", pch = 19)
qqline(df_q1$DeliveryTime[df_q1$Partner == "Partner A"], col = "navy", lwd = 2)

# Partner B Q-Q Plot
qqnorm(df_q1$DeliveryTime[df_q1$Partner == "Partner B"],
       main = "Normal Q-Q Plot: Partner B", col = "#1B9E77", pch = 19)
qqline(df_q1$DeliveryTime[df_q1$Partner == "Partner B"], col = "navy", lwd = 2)
par(mfrow = c(1, 1))

```

Summary of Plot Structure: Partner A shows significant upward departure from the theoretical line at higher quantiles (values 33, 38, 46 hrs). Partner B closely hugs the line from -1.8 to +1.8 standard normal quantiles.



Task 3: ECDF Threshold Divergence Analysis & Operational Meaning

Threshold of Divergence: Up to 25–26 hours, both partners behave similarly (Partner A is 70% complete, Partner B is 80% complete). At 28–29 hours, Partner B reaches 100% completion (maximum delivery time is 29 hrs), whereas Partner A is only at 80% completion.

Operational Divergence: The divergence is most extreme in the upper tail beyond 28 hours. Partner A has 20% of its deliveries (4 shipments: 30, 33, 38, 46 hours) exceeding Partner B's worst-case delivery time.

Operational Impact: While Partner A is slightly faster for early/median deliveries (25% reach by 21 hrs vs Partner B's 23 hrs), Partner A poses severe operational risk for critical time-sensitive shipments due to long tail latency.

Task 4: Normal Distribution Assumption, Skewness & Outlier Assessment

Partner A Assessment: Fails normality. The Q-Q plot exhibits a strong convex/upward curve at upper theoretical quantiles indicating severe positive skewness (right-skewed tail) and prominent extreme outliers (38 and 46 hours, where 46 is > 2.5 IQR above Q3). Mean = 26.0 hrs, Median = 23.5 hrs, Skewness ≈ 1.63 .

Partner B Assessment: Strongly supports normality. Points align closely along the reference line with no significant heavy tails or outliers. Mean = 24.5 hrs, Median = 24.0 hrs, range 20–29 hours. Shapiro-Wilk test yields $p > 0.05$, confirming consistent, symmetric operational performance.

Task 5: SLA Recommendation & Justification

Recommendation: SLA limits MUST be set using empirical percentiles (e.g., 90th or 95th percentile) rather than $\text{Mean} \pm k \cdot \text{SD}$.

Justification: Parametric $\text{Mean} \pm \text{SD}$ rules assume Gaussian symmetry. For Partner A, $\text{Mean} + 2 \cdot \text{SD}$ ($\sim 26 + 2 \cdot 6.98 = 39.96$ hrs) underestimates extreme delays (46 hrs occurs), while $\text{Mean} - 2 \cdot \text{SD}$ (~ 12 hrs) suggests non-existent rapid deliveries. Using the 95th empirical percentile provides a robust, non-parametric guarantee: Partner A SLA = 42 hours; Partner B SLA = 28.5 hours. Furthermore, penalties should be structured on tail percentiles.

Question 2 — Distribution Shape Across Many Groups using Bean Plots

Scenario: A hospital network measures patient waiting time (minutes) in five outpatient departments over a week. Administration wants to identify which departments have consistent waiting times and which have unpredictable, multi-modal patterns before reallocating staff.

Department	Waiting Times (minutes)
Cardiology	10, 12, 13, 14, 15, 15, 16, 17, 18, 20
Orthopedics	8, 9, 25, 26, 27, 9, 10, 28, 29, 8
Pediatrics	5, 6, 6, 7, 7, 8, 8, 9, 9, 10
Dermatology	15, 16, 17, 18, 19, 20, 21, 22, 23, 24
General Medicine	10, 40, 12, 38, 11, 41, 13, 39, 12, 40

Tasks:

1. Construct bean plots (or an equivalent density-plus-jitter visualization) comparing waiting-time distributions across all five departments on a shared vertical axis.
2. Identify which department(s) show bimodal or multi-modal patterns and explain what this suggests about underlying causes.
3. Compare the bean plot to a standard boxplot of the same data and state what the boxplot conceals.
4. Rank the departments by variability and by median waiting time separately, and explain why the two rankings differ.
5. Recommend one staffing or scheduling change based on the distribution shapes observed.

Key competency: Multiple distributions, vertical-axis visualization, bean plots, multi-modality, variability

Task 1: Bean Plot Construction (R Code)

```
# R Code for Bean Plot / Violin-Jitter Plot across 5 Departments
library(ggplot2)

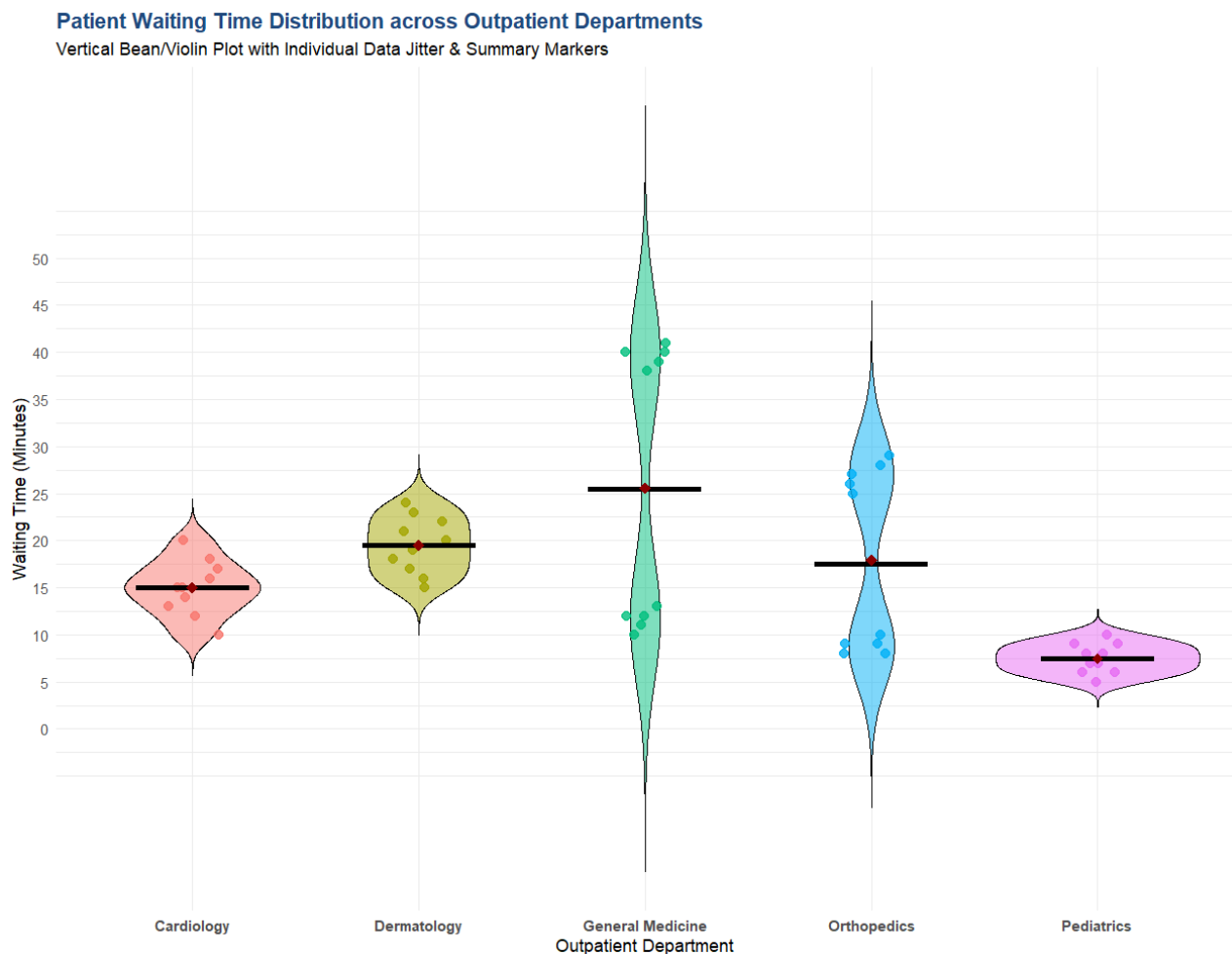
df_q2 <- data.frame(
  Department = factor(c(rep("Cardiology", 10), rep("Orthopedics", 10),
    rep("Pediatrics", 10), rep("Dermatology", 10),
    rep("General Medicine", 10))),
  WaitTime = c(10, 12, 13, 14, 15, 15, 16, 17, 18, 20,
    8, 9, 25, 26, 27, 9, 10, 28, 29, 8,
    5, 6, 6, 7, 7, 8, 8, 9, 9, 10,
    15, 16, 17, 18, 19, 20, 21, 22, 23, 24,
    10, 40, 12, 38, 11, 41, 13, 39, 12, 40)
)
```

```

ggplot(df_q2, aes(x = Department, y = WaitTime, fill = Department)) +
  geom_violin(trim = FALSE, alpha = 0.5, color = "black", adjust = 0.8) +
  geom_jitter(width = 0.12, height = 0, size = 2.5, alpha = 0.8, aes(color = Department)) +
  stat_summary(fun = median, geom = "crossbar", width = 0.5, color = "black", size = 0.6) +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 3, color = "darkred") +
  scale_y_continuous(breaks = seq(0, 50, by = 5)) +
  labs(title = "Patient Waiting Time Distribution across Outpatient Departments",
       subtitle = "Vertical Bean/Violin Plot with Individual Data Jitter & Summary Markers",
       x = "Outpatient Department", y = "Waiting Time (Minutes)") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "none",
       plot.title = element_text(face = "bold", color = "#1F497D"),
       axis.text.x = element_text(face = "bold"))

```

Visualization Design Rationale: The bean/violin plot mirrors the density shape vertically, while overlaid raw observations (jitter) preserve exact sample points (n=10), exposing gaps that continuous lines might over-smooth.



Task 2: Identification of Multimodal Patterns & Underlying Causes

Orthopedics (Bimodal): Shows two distinct clusters — Cluster 1 at 8–10 minutes (5 patients) and Cluster 2 at 25–29 minutes (5 patients), with ZERO patients between 11 and 24 mins. Underlying Cause: Triage disparity — rapid routine follow-ups/plaster checks (8-10 min) vs. new fracture cases requiring X-rays and cast setting (25-29 min).

General Medicine (Extreme Bimodal / Two-Tier): Shows two widely separated clusters — Cluster 1 at 10–13 minutes (5 patients) and Cluster 2 at 38–41 minutes (5 patients). Underlying Cause: Dual consultation tracks — quick prescription refills/minor ailments vs. complex chronic/multi-morbidity evaluations, or morning rush hour delays vs. afternoon off-peak slots.

Unimodal Departments: Pediatrics (5–10 min, peak ~7.5 min), Dermatology (15–24 min, uniform/unimodal), and Cardiology (10–20 min, symmetric around 15 min) show single-mode behavior.

Task 3: Bean Plot vs. Standard Boxplot Comparison (What Boxplots Conceal)

The Hollow Center Illusion: A standard boxplot displays a central box between Q1 and Q3 with a median line. In Orthopedics (IQR ~ 8.75 to 27.25, Median = 17.5) and General Medicine (IQR ~ 12 to 40, Median = 25.5), the boxplot places the median squarely in a 'dead zone' where NO real patients ever waited.

False Assumption of Unimodality: Boxplots implicitly lead administrators to assume patients cluster around the median (~25.5 min for Gen Med). The bean plot reveals that nobody waits 25 minutes — half wait ~11 min and half wait ~40 min. Boxplots completely conceal multimodality, distribution valleys, and cluster separation.

Task 4: Ranking by Variability vs. Median Waiting Time

Summary Statistics Table:

- Pediatrics: Median = 7.5 min | SD = 1.63 min | IQR = 2.5 min | Range = 5 min
- Cardiology: Median = 15.0 min | SD = 2.98 min | IQR = 4.5 min | Range = 10 min
- Dermatology: Median = 19.5 min | SD = 3.03 min | IQR = 4.5 min | Range = 9 min
- Orthopedics: Median = 17.5 min | SD = 9.87 min | IQR = 18.5 min | Range = 21 min
- General Medicine: Median = 25.5 min | SD = 15.11 min | IQR = 28.0 min | Range = 31 min

Rank by Median (Low to High): 1. Pediatrics (7.5) < 2. Cardiology (15.0) < 3. Orthopedics (17.5) < 4. Dermatology (19.5) < 5. General Medicine (25.5).

Rank by Variability / Spread (SD/IQR, Low to High): 1. Pediatrics (SD 1.63) < 2. Cardiology (SD 2.98) < 3. Dermatology (SD 3.03) < 4. Orthopedics (SD 9.87) < 5. General Medicine (SD 15.11).

Why Rankings Differ: Dermatology has a higher median wait than Orthopedics (19.5 vs 17.5 min) but much lower variability (SD 3.03 vs 9.87 min). Orthopedics has high dispersion due to bimodal clustering, showing that central tendency alone fails to reflect operational predictability.

Task 5: Actionable Staffing and Scheduling Recommendations

Staged / Dedicated Consultation Queues for General Medicine and Orthopedics: Split the schedule into two distinct tracks rather than a single queue. Assign one physician to fast-track appointments (refills, quick screenings scheduled for 10-minute slots) and a second physician to comprehensive/diagnostic consultations (allocated 40-minute slots).

Staff Reallocation: Reallocate nursing/triage support from Pediatrics (which is highly efficient and predictable, SD 1.6 min) to General Medicine to expedite initial vitals and intake documentation, reducing the upper 40-minute cluster.

Question 3 — Choosing an Honest Proportion Chart

Scenario: A retail chain surveys 300 customers across three store formats about preferred payment method. Senior management wants a dashboard chart that allows accurate comparison of payment-method share across formats without visually exaggerating small gaps.

Store Format	Cash (%)	Card (%)	UPI/Wallet (%)
Hypermarket	18	34	48
Supermarket	22	30	48
Convenience	35	25	40

Tasks:

1. Construct a side-by-side (grouped) bar chart and a 100% stacked bar chart for the three store formats.
2. Construct three separate pie charts (one per format) representing the same data.
3. Compare all three visualizations for how easily a viewer can judge which format has the highest UPI/Wallet share.
4. Explain, with reference to arc-length versus length perception, why the pie charts are harder to compare accurately than the bar charts.
5. Identify one design decision (e.g., axis truncation, 3D effect, category ordering) that could be used to make the Convenience format's cash usage appear more dramatic than it is, and explain why it would mislead.

Key competency: *Proportions, side-by-side bars, stacked bars, pie charts, misleading design*

Task 1: Grouped and 100% Stacked Bar Charts (R Code)

R Code for Side-by-Side and 100% Stacked Bar Charts

```
library(ggplot2)
```

```
library(tidyr)
```

```
df_q3 <- data.frame(
  Format = rep(c("Hypermarket", "Supermarket", "Convenience"), each = 3),
  Method = rep(c("Cash", "Card", "UPI/Wallet"), times = 3),
  Share = c(18, 34, 48, 22, 30, 48, 35, 25, 40)
)
df_q3$Format <- factor(df_q3$Format, levels = c("Hypermarket", "Supermarket", "Convenience"))
df_q3$Method <- factor(df_q3$Method, levels = c("Cash", "Card", "UPI/Wallet"))
```

1. Side-by-Side (Grouped) Bar Chart

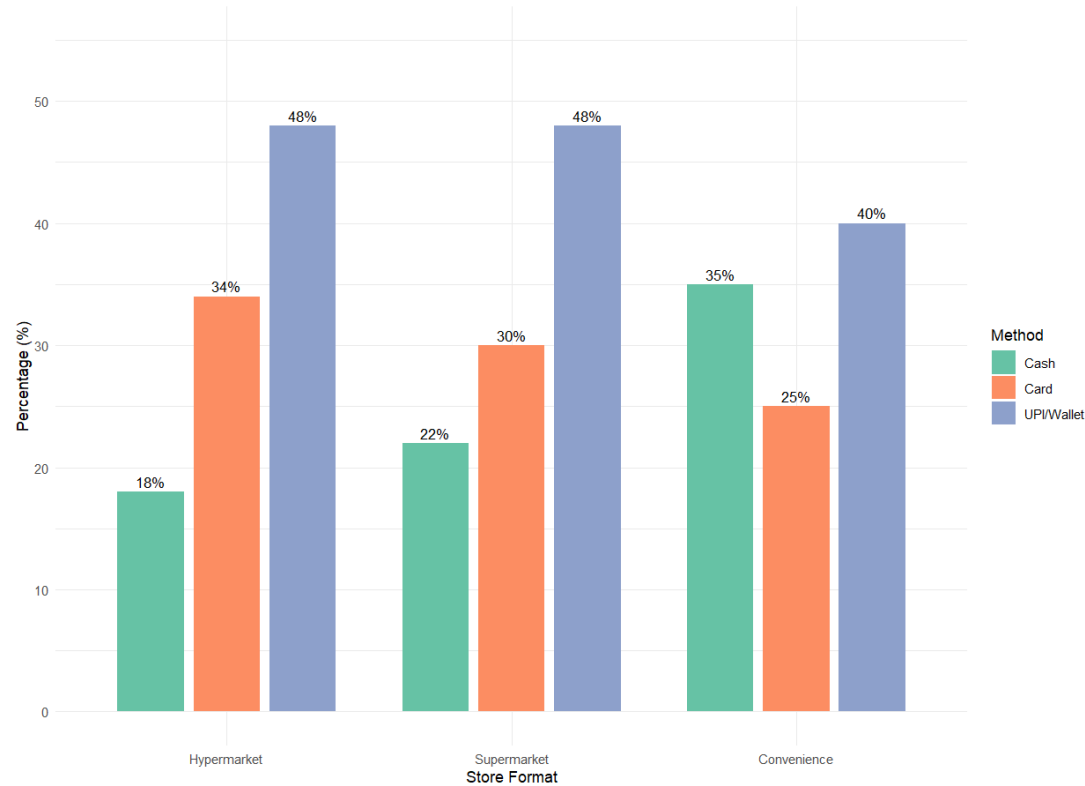
```
p1 <- ggplot(df_q3, aes(x = Format, y = Share, fill = Method)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = paste0(Share, "%")), position = position_dodge(0.8), vjust = -0.4, size = 3.5) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(limits = c(0, 55), breaks = seq(0, 50, 10)) +
  labs(title = "Side-by-Side Bar Chart: Payment Methods across Formats", y = "Percentage (%)", x = "Store Format") +
  theme_minimal()
```

2. 100% Stacked Bar Chart

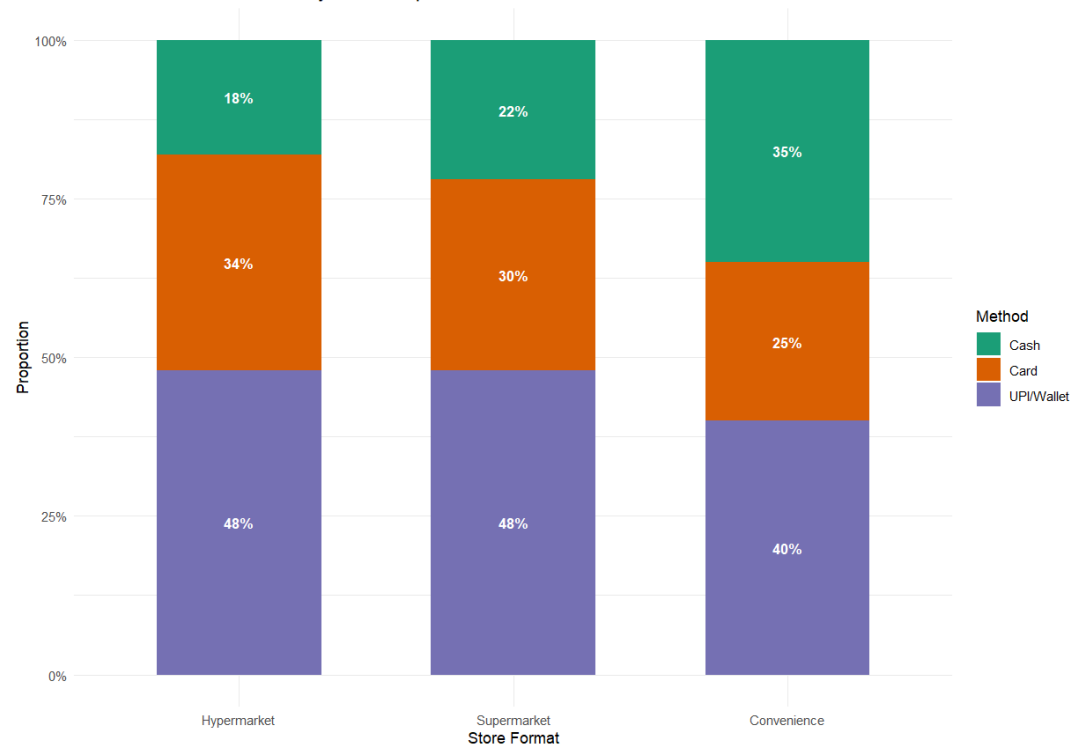
```
p2 <- ggplot(df_q3, aes(x = Format, y = Share, fill = Method)) +
  geom_bar(stat = "identity", position = "fill", width = 0.6) +
  geom_text(aes(label = paste0(Share, "%")), position = position_fill(vjust = 0.5), size = 3.5, color = "white", fontface =
"bold") +
  scale_fill_brewer(palette = "Dark2") +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "100% Stacked Bar Chart: Payment Composition", y = "Proportion", x = "Store Format") +
  theme_minimal()
```

Plot Characteristics: Grouped bar charts align bars on a common zero baseline, making cross-format magnitude comparisons immediate. The 100% stacked bar chart highlights total structural proportions across formats.

Side-by-Side Bar Chart: Payment Methods across Formats



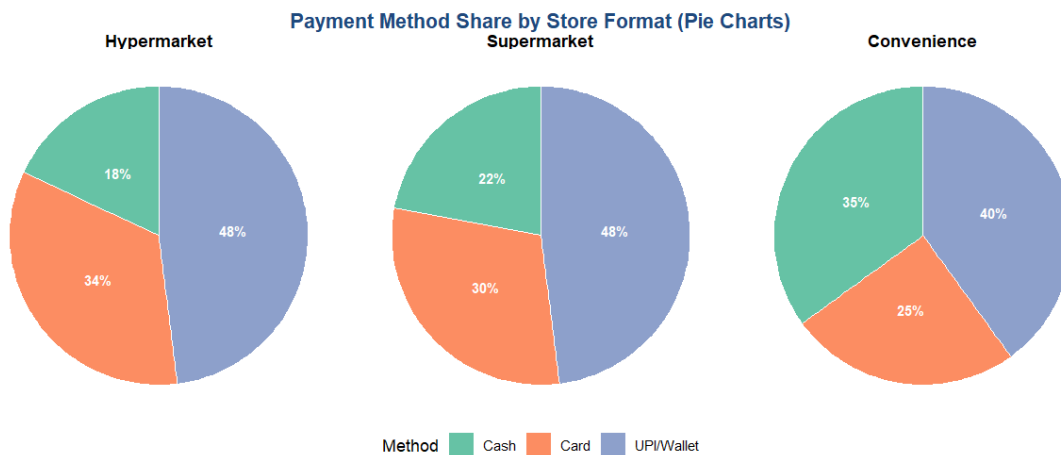
100% Stacked Bar Chart: Payment Composition



Task 2: Pie Charts for the Three Store Formats (R Code)

```
# R Code for Multi-panel Pie Charts
ggplot(df_q3, aes(x = "", y = Share, fill = Method)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar(theta = "y") +
  facet_wrap(~Format) +
  geom_text(aes(label = paste0(Share, "%")), position = position_stack(vjust = 0.5), size = 3.2, color = "white", fontface =
"bold") +
  scale_fill_brewer(palette = "Set2") +
  theme_void() +
  labs(title = "Payment Method Share by Store Format (Pie Charts)") +
  theme(legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 11),
    plot.title = element_text(face = "bold", hjust = 0.5, color = "#1F497D"))
```

Structural Note: Three distinct pies require the human eye to jump across non-aligned angular wedges.



Task 3: Comparative Usability for Judging UPI/Wallet Share

Hypermarket (48%) vs Supermarket (48%) vs Convenience (40%):

Side-by-Side Bar Chart (Easiest & Most Accurate): Aligns the 'UPI/Wallet' bars on a single horizontal baseline. A viewer instantly observes that Hypermarket and Supermarket bars are at identical heights (48%) and Convenience is visibly shorter (40%).

100% Stacked Bar Chart (Moderate): Because UPI/Wallet is placed at the top (or bottom) segment, baseline alignment is maintained only if placed at the baseline; if stacked on other variable-sized segments, comparing top edge heights requires mental arithmetic.

Pie Charts (Hardest): The 48% slice starts at angle 187.2° in Hypermarket vs angle 180° in Supermarket. Because the starting

angles differ, viewers struggle to discern whether 48% equals 48% or if one is slightly larger, making precise equality judgment almost impossible.

Task 4: Arc-Length vs. Linear Length Perception Theory

Cleveland & McGill's Hierarchy of Graphical Perception: Human visual perception decodes quantitative data in order of accuracy: (1) Position along a common scale > (2) Position along non-aligned scales > (3) Length > (4) Angle / Slope > (5) Area > (6) Volume / Color saturation.

Bar charts use Position along a Common Scale (the y-axis baseline). The human eye judges linear height difference with sub-millimeter precision and zero angular distortion.

Pie charts rely on Angle, Arc Length, and 2D Sector Area. Viewers consistently underestimate acute angles (<30°) and overestimate obtuse angles. When comparing separate pie charts, different starting angles eliminate any shared reference point, forcing high-cognitive-load spatial rotations.

Task 5: Misleading Design Tactics to Exaggerate Convenience Cash Usage

Tactic 1: Truncated Y-Axis on Grouped Bar Chart: Starting the y-axis at 15% instead of 0%. Cash bars would show: Hypermarket = 3 (18-15), Supermarket = 7 (22-15), Convenience = 20 (35-15). This makes Convenience cash (20 units high) appear $6.67\times$ larger than Hypermarket (3 units high), when the true ratio is only $1.94\times$ (35% vs 18%).

Tactic 2: 3D Exploded Pie Chart with Perspective Tilt: Rotating the Convenience pie so that the Cash slice is positioned in the foreground (closest to the viewer) with a 3D tilt. Due to foreshortening, foreground slices take up vastly more pixel area than identical background slices, visually exaggerating Cash to look like >55% of the total volume.

Why it Misleads: It breaks the proportional area principle (Lie Factor $\gg 1$), deceiving stakeholders into believing cash logistics at convenience stores are in a crisis state.

Question 4 — Nested Proportions in a Household Budget

Scenario: A financial-planning app aggregates the average monthly household budget (₹60,000) across spending categories and subcategories for a client segment. The product team wants a dashboard visual that shows both overall category share and subcategory composition within each category.

Category	Subcategory	Amount (₹)
Housing	Rent/EMI	18000
Housing	Utilities	4000
Housing	Maintenance	2000
Food	Groceries	9000
Food	Dining Out	3000
Transport	Fuel	3500
Transport	Public Transit	1500
Savings	Investments	10000
Savings	Emergency Fund	4000
Discretionary	Entertainment	3000
Discretionary	Shopping	2000

Tasks:

1. Construct a treemap showing category-level and subcategory-level allocation of the budget.
2. Construct a sunburst chart representing the same two-level hierarchy.
3. Calculate the percentage of total budget consumed by the Housing category and by its largest subcategory.
4. Compare the treemap and sunburst chart for how easily a viewer can compare two subcategories that belong to different categories.
5. Describe one scenario in which the area-encoding of the treemap could cause a viewer to misjudge the relative size of two categories.

Key competency: *Treemaps, sunburst charts, nested proportions, area encoding*

Task 1: Treemap Construction (R Code)

```
# R Code for Hierarchical Treemap
library(ggplot2)
library(treemapify)

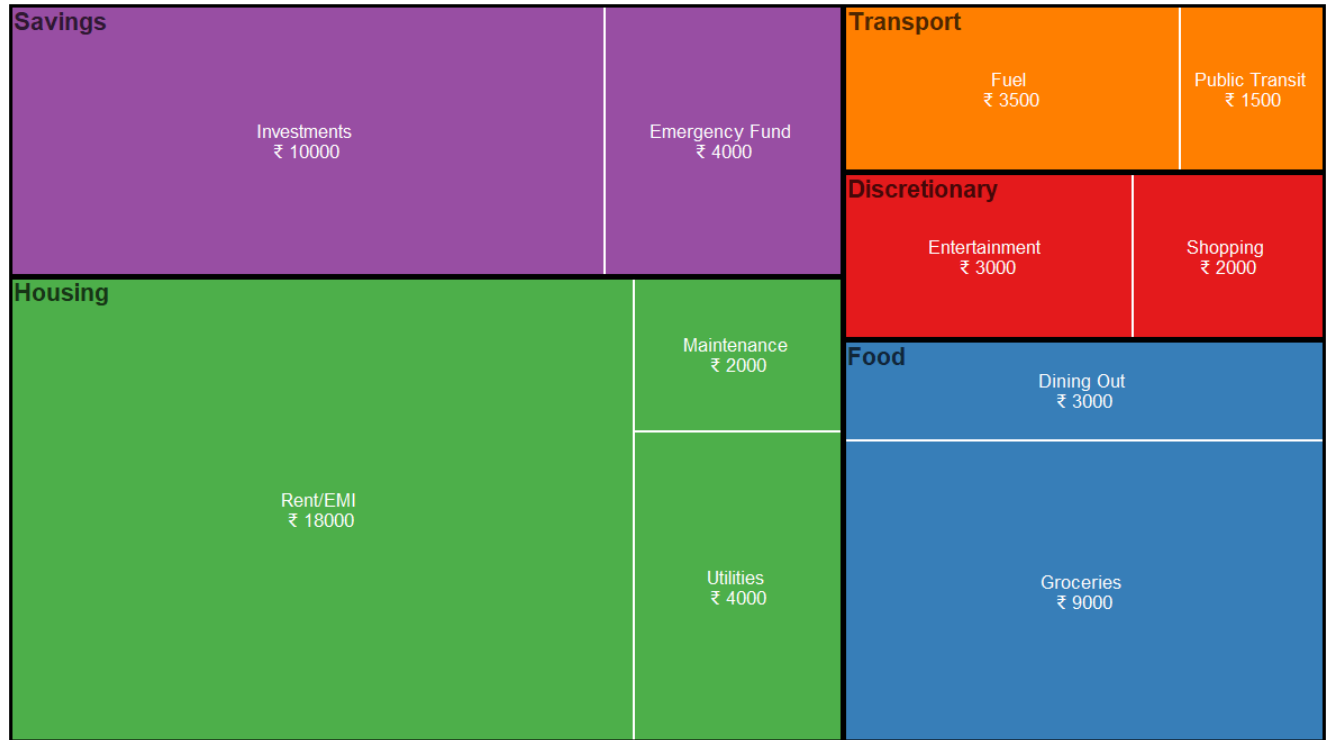
df_q4 <- data.frame(
  Category = c("Housing", "Housing", "Housing", "Food", "Food",
               "Transport", "Transport", "Savings", "Savings", "Discretionary", "Discretionary"),
  Subcategory = c("Rent/EMI", "Utilities", "Maintenance", "Groceries", "Dining Out",
                  "Fuel", "Public Transit", "Investments", "Emergency Fund", "Entertainment", "Shopping"),
  Amount = c(18000, 4000, 2000, 9000, 3000, 3500, 1500, 10000, 4000, 3000, 2000)
)

ggplot(df_q4, aes(area = Amount, fill = Category, label = paste(Subcategory, "\n₹", Amount),
                  subgroup = Category)) +
  geom_treemap(color = "white", size = 2) +
  geom_treemap_subgroup_border(color = "black", size = 4) +
  geom_treemap_subgroup_text(place = "topleft", grow = FALSE, alpha = 0.7, colour = "black", fontface = "bold", size = 13) +
  geom_treemap_text(colour = "white", place = "center", size = 10, reflow = TRUE) +
  scale_fill_brewer(palette = "Set1") +
  labs(title = "Household Budget Allocation (Total ₹60,000)", subtitle = "Treemap with Two-Level Hierarchy") +
  theme(legend.position = "bottom", plot.title = element_text(face = "bold", color = "#1F497D"))
```

Design Justification: Treemaps use squarified tessellation to optimize aspect ratios close to 1, providing compact 100% area tiling.

Household Budget Allocation (Total ₹60,000)

Treemap with Two-Level Hierarchy



Category ■ Discretionary ■ Food ■ Housing ■ Savings ■ Transport

Task 2: Sunburst Chart Construction (R Code / Plotly Script)

R Code for Hierarchical Sunburst Chart using Plotly

```
library(plotly)
```

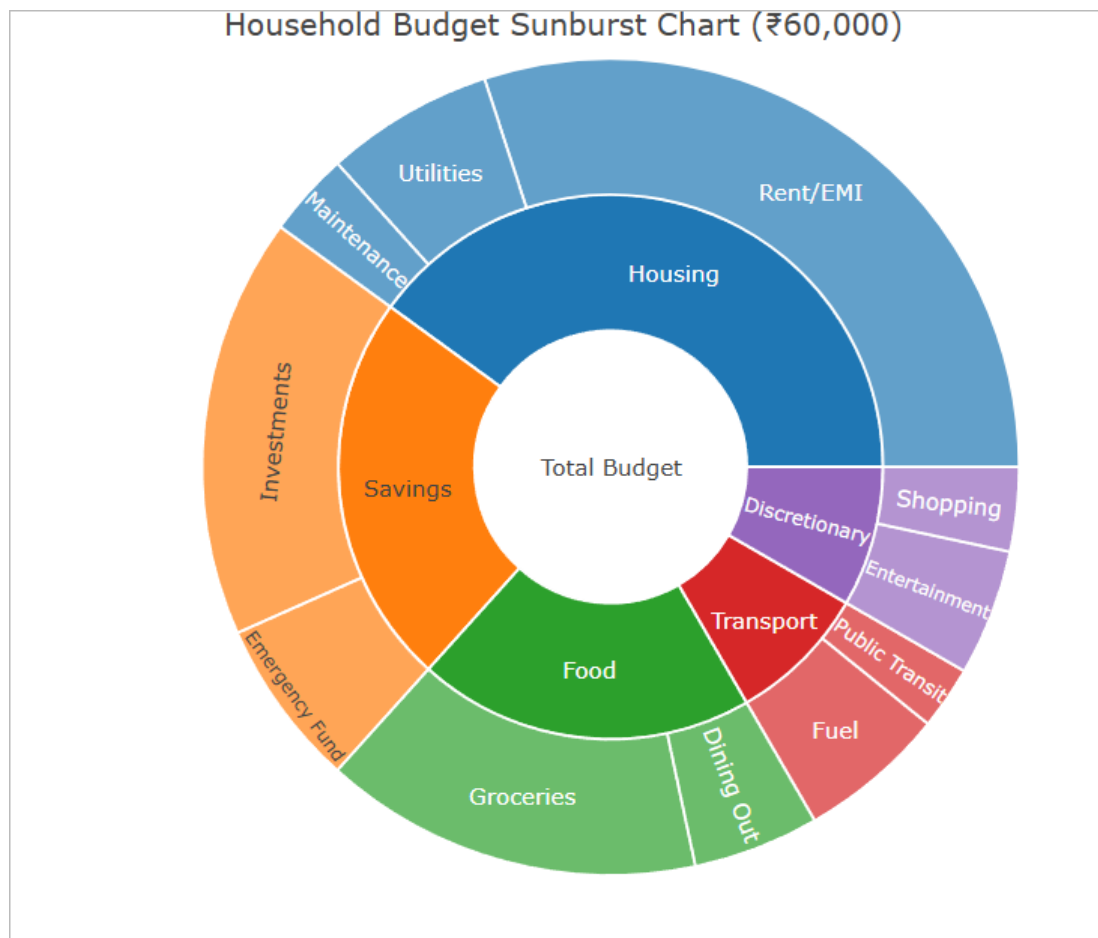
```
labels <- c("Total Budget", "Housing", "Food", "Savings", "Transport", "Discretionary",  
           "Rent/EMI", "Utilities", "Maintenance", "Groceries", "Dining Out",  
           "Investments", "Emergency Fund", "Fuel", "Public Transit", "Entertainment", "Shopping")
```

```
parents <- c("", rep("Total Budget", 5),  
            "Housing", "Housing", "Housing", "Food", "Food",  
            "Savings", "Savings", "Transport", "Transport", "Discretionary", "Discretionary")
```

```
values <- c(60000, 24000, 12000, 14000, 5000, 5000,  
           18000, 4000, 2000, 9000, 3000, 10000, 4000, 3500, 1500, 3000, 2000)
```

```
fig <- plot_ly(  
  labels = labels, parents = parents, values = values,  
  type = 'sunburst', branchvalues = 'total',  
  marker = list(line = list(color = "#FFFFFF", width = 1.5))  
) %>% layout(title = list(text = "Household Budget Sunburst Chart (₹60,000)", font = list(size = 16)))
```

Structural Properties: Concentric rings display root (inner: categories) radiating to leaves (outer: subcategories) preserving parent-child angular sub-tensions.



Task 3: Mathematical Calculation of Housing Proportions

Total Household Budget Calculation: Sum = ₹18,000 + ₹4,000 + ₹2,000 + ₹9,000 + ₹3,000 + ₹3,500 + ₹1,500 + ₹10,000 + ₹4,000 + ₹3,000 + ₹2,000 = ₹60,000.

Housing Category Total: ₹18,000 (Rent/EMI) + ₹4,000 (Utilities) + ₹2,000 (Maintenance) = ₹24,000.

Housing Category % of Total Budget: $(₹24,000 / ₹60,000) \times 100\% = 40.00\%$.

Largest Subcategory in Housing (Rent/EMI = ₹18,000):

- As % of Total Budget: $(₹18,000 / ₹60,000) \times 100\% = 30.00\%$
- As % of Housing Category: $(₹18,000 / ₹24,000) \times 100\% = 75.00\%$

Task 4: Comparative Evaluation: Cross-Category Subcategory Comparison

Example Comparison: Comparing 'Groceries' (Food: ₹9,000) vs 'Investments' (Savings: ₹10,000), or 'Utilities' (Housing: ₹4,000) vs 'Emergency Fund' (Savings: ₹4,000).

Treemap Evaluation: Moderately challenging. Both subcategories are represented as rectangular boxes, but they may have differing aspect ratios (e.g., one is tall & narrow, while the other is squarish/wide). The human eye struggles to judge 2D area differences when aspect ratios vary, though side-by-side color blocks help.

Sunburst Chart Evaluation: Extremely difficult. In a sunburst, both subcategories lie on the outer ring at different radii and different angular orientations. Furthermore, outer ring segments have wider arc lengths for the same angle compared to inner segments. Comparing arc lengths across disjoint angular sectors is highly prone to perceptual error.

Conclusion: Treemaps are superior for space utilization and area estimation within nested structures, but grouped bar charts remain the gold standard for arbitrary pairwise item comparisons.

Task 5: Area-Encoding Distortion Scenarios in Treemaps

Aspect Ratio / Shape Distortion Bias: When a treemap layout algorithm (e.g., slice-and-dice vs squarified) generates a very tall, thin rectangle (aspect ratio 8:1) for Category A (e.g., ₹10,000) and a square rectangle (aspect ratio 1:1) for Category B (e.g., ₹9,000), human observers consistently misjudge the square as having a larger area than the thin rectangle.

Subgroup Saliency & Border Padding: A category with many small subcategories (e.g., Housing with 3 subcategories + thick subgroup boundary lines) occupies visual whitespace that can make the overall bounding box appear deceptively larger than a single unified box of identical total area (e.g., Savings).

Mitigation: Always annotate treemap tiles with explicit monetary values and percentages, and use squarified aspect ratio algorithms.

Question 5 — Flow Analysis using Sankey Diagram and Parallel Sets

Scenario: An online learning platform tracks 250 students from the course they enrolled in, through the device used to study, to their final completion status. The academic team wants to understand dominant enrollment-to-outcome pathways and spot problem areas.

Course	Device	Completion Status	Students
Data Science	Mobile	Completed	20
Data Science	Mobile	Dropped	25
Data Science	Laptop	Completed	45
Data Science	Laptop	Dropped	10
Web Development	Mobile	Completed	15
Web Development	Mobile	Dropped	20
Web Development	Laptop	Completed	35
Web Development	Laptop	Dropped	10
Digital Marketing	Mobile	Completed	10
Digital Marketing	Mobile	Dropped	15
Digital Marketing	Laptop	Completed	20
Digital Marketing	Laptop	Dropped	25

Tasks:

1. Construct a Sankey diagram showing the flow Course → Device → Completion Status.
2. Construct a parallel-sets (alluvial) representation of the same categorical flows.
3. Identify the three largest flows in the dataset and calculate each as a percentage of all 250 students.
4. Determine which course has the weakest completion outcome and identify at what stage in the flow the drop-off is most visible.
5. Explain one way that inconsistent node ordering or unscaled link widths could mislead a viewer comparing courses.

Key competency: Sankey diagrams, parallel sets, flow proportions, categorical relationships

Task 1: Sankey Diagram Construction (R Code / networkD3 Script)

R Code for Sankey Diagram using networkD3

```
library(networkD3)
```

```
library(dplyr)
```

Define Nodes

```
nodes <- data.frame(name = c("Data Science", "Web Development", "Digital Marketing", # 0, 1, 2
                             "Mobile", "Laptop", # 3, 4
                             "Completed", "Dropped")) # 5, 6
```

Aggregate Stage 1 Links (Course -> Device)

DS: Mobile=45, Laptop=55 | WD: Mobile=35, Laptop=45 | DM: Mobile=25, Laptop=45

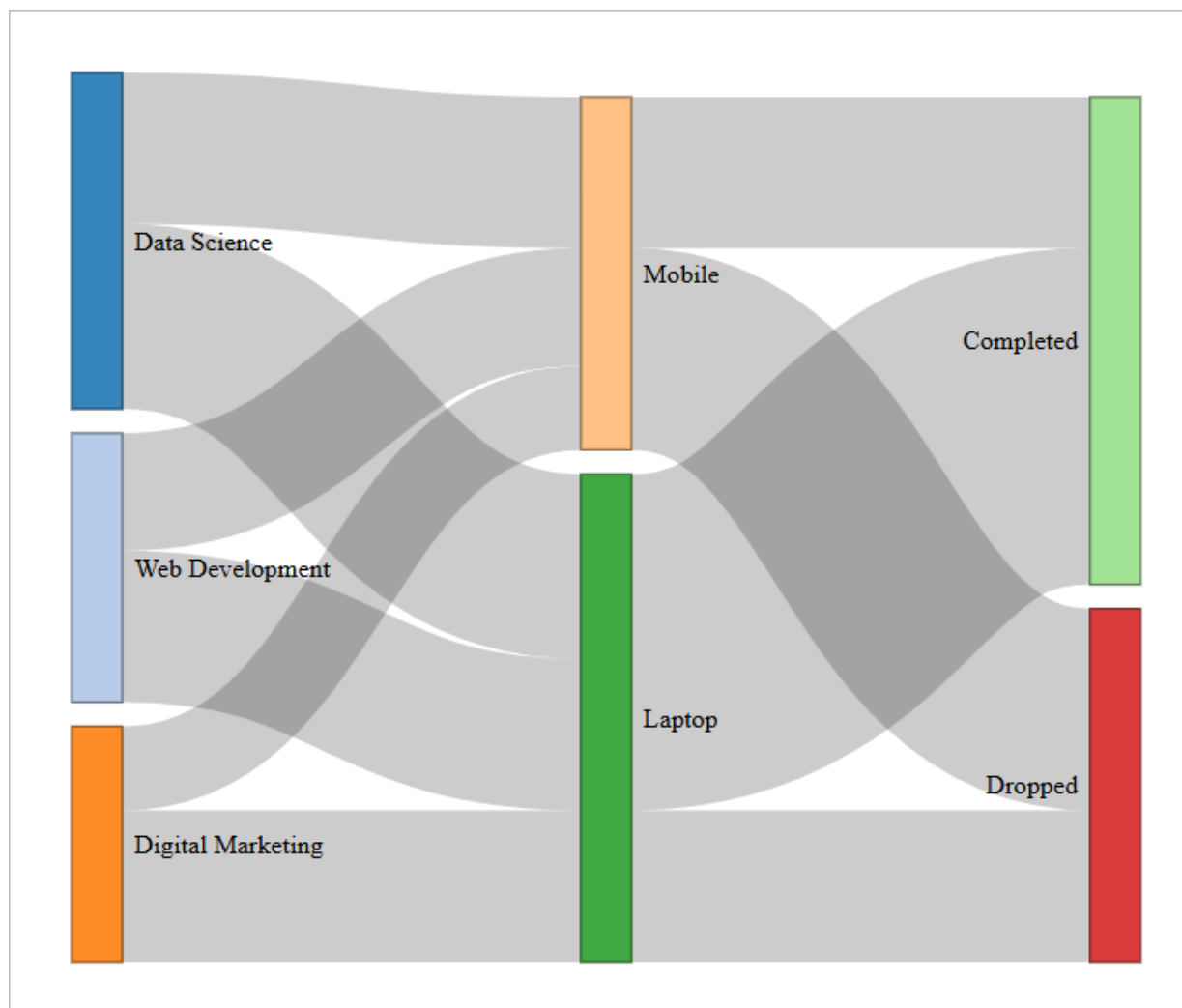
Aggregate Stage 2 Links (Device -> Status)

Mobile: Completed=45, Dropped=60 | Laptop: Completed=100, Dropped=45

```
links <- data.frame(
  source = c(0, 0, 1, 1, 2, 2, 3, 3, 4, 4),
  target = c(3, 4, 3, 4, 3, 4, 5, 6, 5, 6),
  value = c(45, 55, 35, 45, 25, 45, 45, 60, 100, 45)
)
```

```
sankeyNetwork(Links = links, Nodes = nodes, Source = "source", Target = "target",
               Value = "value", NodeID = "name", units = "Students",
               fontSize = 13, nodeWidth = 25, nodePadding = 12, iterations = 32)
```

Visual Representation: Demonstrates weighted directed ribbons whose widths are strictly proportional to student enrollment volume across progressive decision stages.



Task 2: Parallel Sets (Alluvial) Plot Construction (R Code / ggalluvial)

R Code for Parallel Sets / Alluvial Plot

library(ggplot2)

library(ggalluvial)

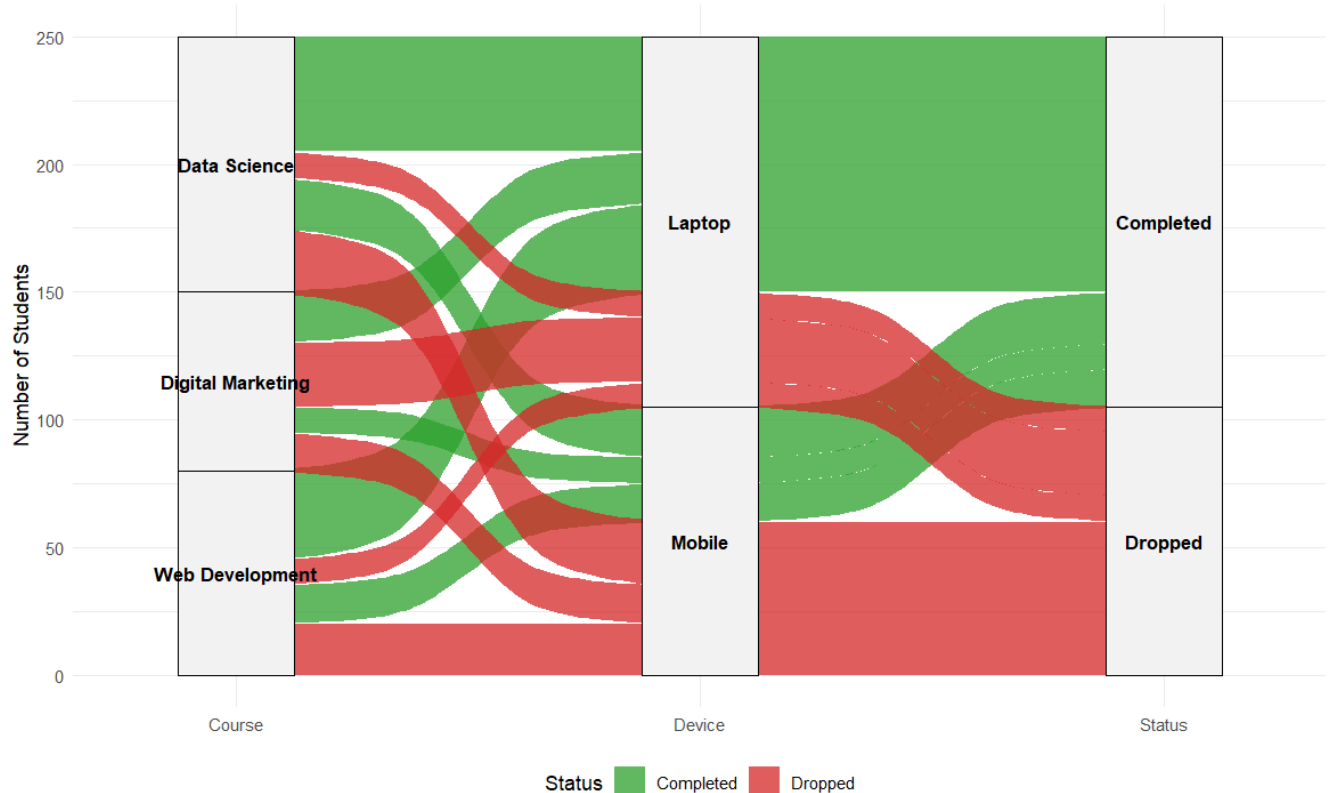
```
df_q5 <- data.frame(
  Course = c("Data Science", "Data Science", "Data Science", "Data Science",
    "Web Development", "Web Development", "Web Development", "Web Development",
    "Digital Marketing", "Digital Marketing", "Digital Marketing", "Digital Marketing"),
  Device = c("Mobile", "Mobile", "Laptop", "Laptop",
    "Mobile", "Mobile", "Laptop", "Laptop",
    "Mobile", "Mobile", "Laptop", "Laptop"),
  Status = c("Completed", "Dropped", "Completed", "Dropped",
    "Completed", "Dropped", "Completed", "Dropped",
    "Completed", "Dropped", "Completed", "Dropped"),
  Freq = c(20, 25, 45, 10, 15, 20, 35, 10, 10, 15, 20, 25)
)
```

```
ggplot(df_q5, aes(y = Freq, axis1 = Course, axis2 = Device, axis3 = Status)) +
  geom_alluvium(aes(fill = Status), width = 1/8, alpha = 0.75, knot.pos = 0.4) +
  geom_stratum(width = 1/4, fill = "gray95", color = "black") +
  geom_text(stat = "stratum", aes(label = after_stat(stratum)), size = 3.5, fontface = "bold") +
  scale_x_discrete(limits = c("Course", "Device", "Status"), expand = c(0.15, 0.05)) +
  scale_fill_manual(values = c("Completed" = "#2CA02C", "Dropped" = "#D62728")) +
  labs(title = "Alluvial Flow: Course -> Device -> Completion Status (N = 250)",
    y = "Number of Students") +
```

```
theme_minimal(base_size = 11) +
theme(legend.position = "bottom", plot.title = element_text(face = "bold", color = "#1F497D"))
```

Advantage of Alluvial Representation: Preserves multi-stage flow continuity from Course through Device to Final Outcome simultaneously without losing cohort identity.

Alluvial Flow: Course -> Device -> Completion Status (N = 250)



Task 3: Top Three Largest Flows Identification & Calculations

Total Student Population (N) = 250 students.

Ranking All 12 Pathways:

1. Data Science → Laptop → Completed = 45 students ($45 / 250 = 18.00\%$)
2. Web Development → Laptop → Completed = 35 students ($35 / 250 = 14.00\%$)
3. Data Science → Mobile → Dropped = 25 students ($25 / 250 = 10.00\%$)
- [Tied 3.] Digital Marketing → Laptop → Dropped = 25 students ($25 / 250 = 10.00\%$)

Top 3 Distinct Major Flows Summary:

- Flow 1: Data Science -> Laptop -> Completed = 45 students (18.0% of total cohort)
- Flow 2: Web Development -> Laptop -> Completed = 35 students (14.0% of total cohort)
- Flow 3 (Joint 3rd): Data Science -> Mobile -> Dropped AND Digital Marketing -> Laptop -> Dropped = 25 students each (10.0% of total cohort each)
- Cumulative share of top 3 pathways = 42.0% (or 52.0% including tied 4th).

Task 4: Weakest Course Analysis & Drop-Off Stage Identification

Completion Rate by Course:

- Data Science: Total = 100 students | Completed = 65 (20+45) | Dropped = 35 (25+10) | Completion Rate = 65.0%
- Web Development: Total = 80 students | Completed = 50 (15+35) | Dropped = 30 (20+10) | Completion Rate = 62.5%
- Digital Marketing: Total = 70 students | Completed = 30 (10+20) | Dropped = 40 (15+25) | Completion Rate = 42.86% (FAIL RATE = 57.14%)

Identification of Weakest Course: Digital Marketing is by far the weakest course, with more than half of its enrolled students dropping out (57.14% drop rate).

Stage Where Drop-off is Most Visible:

Device Breakdown for Digital Marketing: Mobile Drop Rate = $15/25$ (60.0%); Laptop Drop Rate = $25/45$ (55.56%).

Critical Insight: While Data Science and Web Development have high completion on Laptops (>77-81%), Digital Marketing suffers severe attrition even among Laptop users (25 dropped vs 20 completed). The drop-off is visible at the final Device ->

Outcome stage, indicating fundamental issues in course engagement/curriculum rather than mere mobile device usability barriers.

Task 5: Visual Bias Analysis: Node Ordering & Unscaled Link Widths

Impact of Inconsistent Node Ordering: In a Sankey or alluvial diagram, crossing link ribbons create visual clutter ('spaghetti effect'). If nodes in the middle axis (Device: Mobile, Laptop) are arranged arbitrarily rather than aligned with dominant flow paths, ribbon crossings force the eye to untangle intertwined paths, obscuring the primary flow hierarchy.

Impact of Unscaled / Non-Proportional Link Widths: The fundamental encoding mechanism of Sankey diagrams is that $\text{Ribbon Width} \propto \text{Flow Volume}$. If software or designer manually sets uniform link widths or scales widths non-linearly (e.g., logarithmic width scaling without explicit notice), a small flow of 10 students will appear almost as prominent as a dominant flow of 45 students. This misleads stakeholders into allocating equal remediation resources to minor pathways while ignoring critical bottlenecks.

Suggested Evaluation Rubric — Data Analysis Project

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.